4	To teach the fundamental techniques and principles in achieving big data analytics with scalability and streaming capability.
5	To introduce to the students several types of big data like social media, web graphs and data streams.
6	To enable students to have skills that will help them to solve complex real-world problems in decision
Course Outcomes: After successful completion of the course students will be able to:	
1	Explain the motivation for big data systems and identify the main sources of Big Data in the real world.
2	Demonstrate an ability to use frameworks like Hadoop, NOSQL to efficiently store, retrieve and process Big Data for Analytics.
3	Implement several Data Intensive tasks using the Map Reduce Paradigm.
4	Apply several newer algorithms for Clustering Classifying and finding associations in Big Data.
5	Design algorithms to analyze Big data like streams, Web Graphs and Social Media data.
6	Design and implement successful Recommendation engines for enterprises.

Syllabus

Module No.		Content	Hours
1.		**Introduction to Big Data**	05
	1.1	Introduction to Big Data, Big Data characteristics, types of Big Data.	
	1.2	Traditional vs. Big Data business approach, Big Data Challenges.	
	1.3	Examples of Big Data in Real Life, Big Data Applications.	
		Self Learning Topics: Identification of Big Data applications and its solutions.	

2.		**Introduction to Big Data Frameworks**	**06**
	2.1	What is Hadoop? Core Hadoop Components;	
	2.2	Hadoop Ecosystem; Working with Apache Spark	
	2.3	What is NoSQL? NoSQL data architecture patterns: Key value stores, Graph stores, Column family stores, Document stores, MongoDB and its applications.	
		Self Learning Topics:HDFS vs GFS, MongoDB vs other NoSQL system, Implementation of Apache Spark.	
3.		**MapReduce Paradigm**	**07**
	3.1	MapReduce,The Map Tasks, Grouping by Key, The Reduce Tasks, Combiners, Details of MapReduce Execution, Coping With Node Failures.	
	3.2	Algorithms Using MapReduce: Matrix Vector Multiplication by MapReduce, Relational-Algebra Operations, Computing Selections by MapReduce, Computing Projections by MapReduce, Union, Intersection, and Difference by MapReduce, Computing Natural Join by MapReduce, Grouping and Aggregation by MapReduce,	
	3.3	Matrix Multiplication, Matrix Multiplication with One MapReduce Step and Two Step MapReduce.	
	3.4	Illustrating use of MapReduce with use of real life databases and applications.	
		Self Learning Topics: Implementation of MapReduce algorithms like Word count, Matrix-Vector and Matrix Matrix algorithm.	
4.		**Mining Big Data Streams**	**07**
	4.1	The Stream Data Model:A DataStream-Management System, Examples of Stream Sources.	

	4.2	Stream Queries, Issues in Stream Processing. Sampling Data in a Stream : Sampling Techniques.	
	4.3	Filtering Streams: The Bloom Filter Counting Distinct Elements in a Stream : The Count-Distinct Problem.	
	4.4	The Flajolet-Martin Algorithm, Combining Estimates, Space Requirements . Counting Ones in a Window: The Cost of Exact Counts.	
	4.5	The Datar-Gionis-Indyk, Motwani Algorithm, Query Answering in the DGIM Algorithm.	
		Self Learning Topics: Streaming services like Apache Kafka/Amazon Kinesis/Google Cloud DataFlow. Standard spark streaming library.Integration with IOT devices to capture real time stream data.	
5.		**Big Data Mining Algorithms**	**07**
	5.1	Frequent Pattern Mining : Handling Larger Datasets in Main Memory.	
	5.2	Basic Algorithm of Park, Chen, and Yu, Multistage PCY algorithm.	
	5.3	The SON Algorithm and MapReduce.	
	5.4	Clustering Algorithms:CURE Algorithm.,Canopy Clustering, Clustering with MapReduce.	
	5.5	Classification Algorithms: Overview SVM classifiers, Programme Parallel SVM.	
		Self Learning Topics: Standard libraries included with spark like graphX, MLlib.	
6.		**Link Analysis,Mining Social- Network Graphs and Recommendation Engines**	**07**

	6.1	PageRank Definition, Structure of the web, dead ends, Using Page rank in a search engine, Efficient computation of Page Rank: PageRank Iteration Using MapReduce, Topic sensitive Page Rank, link Spam, Hubs and Authorities, HITS Algorithm.	
	6.2	Social Networks as Graphs, Types , Clustering of Social Network Graphs, Direct Discovery of Communities, Counting triangles using Map Reduce.	
	6.3	A Model for Recommendation Systems, Content-Based Recommendations, Collaborative Filtering.	
		Self Learning Topics: Sample applications like social media feeds, multiplayer game interactions, retail industry, financial data analysis. Use cases like location data, real-time stock trades, log monitoring etc.	
		TOTAL	**39**

Textbooks:	
1	Anand Rajaraman and Jeff Ullman "Mining of Massive Datasets", Cambridge University Press. 2. Alex Holmes "Hadoop in Practice", Manning Press, Dreamtech Press.
2	Professional NoSQL Paperback, by Shashank Tiwari, Dreamtech Press
3	Rajkumar Buyya, ,Rodrigo N. Calheiros and Amir Vahid Dastjerdi, "Big Data Principles and Paradigms", Morgan Kaufmann
Reference Books:	
1	Analytics in a Big Data World: The Essential Guide to Data Science and its Applications, Bart Baesens , WILEY Big Data Series.
2	Big Data Analytics with R and Hadoop by Vignesh Prajapati Paperback, Packt Publishing Limited 3. Hadoop: The Definitive Guide by Tom White, O'Reilly Publications